CIS 227 LAB REPORT

Server Administration & Network Security

Lab Project 2 — Install Roles and Features

BACKGROUND	
Course:	CIS 227 - Server Administration & Network Security
Term:	Spring 2026
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Lab Title:	Lab Project 2 - Install Roles and Features
Week / Module Reference:	Week 2 / Windows Server 2016 Roles and Features
Lab Type:	Hands-On Virtual Machine Lab
Date Completed:	2026-04-20
Time Spent:	Approximately 2-3 hours

ENVIRONMENT

Platforms, Systems, and Interfaces Used:

- Primary host platform: Oracle VM VirtualBox on a Windows desktop environment.
- Guest system: CIS227-WS2016-Datacenter running Windows Server 2016 Datacenter (Desktop Experience).
- VM configuration: 4096 MB RAM, 2 virtual CPUs, 64 GB virtual disk, and NAT networking.
- Interfaces used during the work included Server Manager, Windows PowerShell, Internet Explorer, and File Explorer.

Work Objective:

- Install and validate Windows Server 2016 roles and features on the lab virtual machine.
- Use Server Manager for GUI-based installation and PowerShell for direct role, service, and network administration.
- Confirm that installed services function correctly through local testing.

Work Summary:

- Installed DHCP Server and Windows Server Backup through Server Manager.
- Installed IIS Web Server and File Server through Windows PowerShell.
- Created and verified a local SMB shared folder at C:\Shared.
- Tested IIS locally by loading http://localhost.
- Inspected the network configuration, replaced DHCP client addressing with a static address, corrected DNS settings, and verified connectivity afterward.

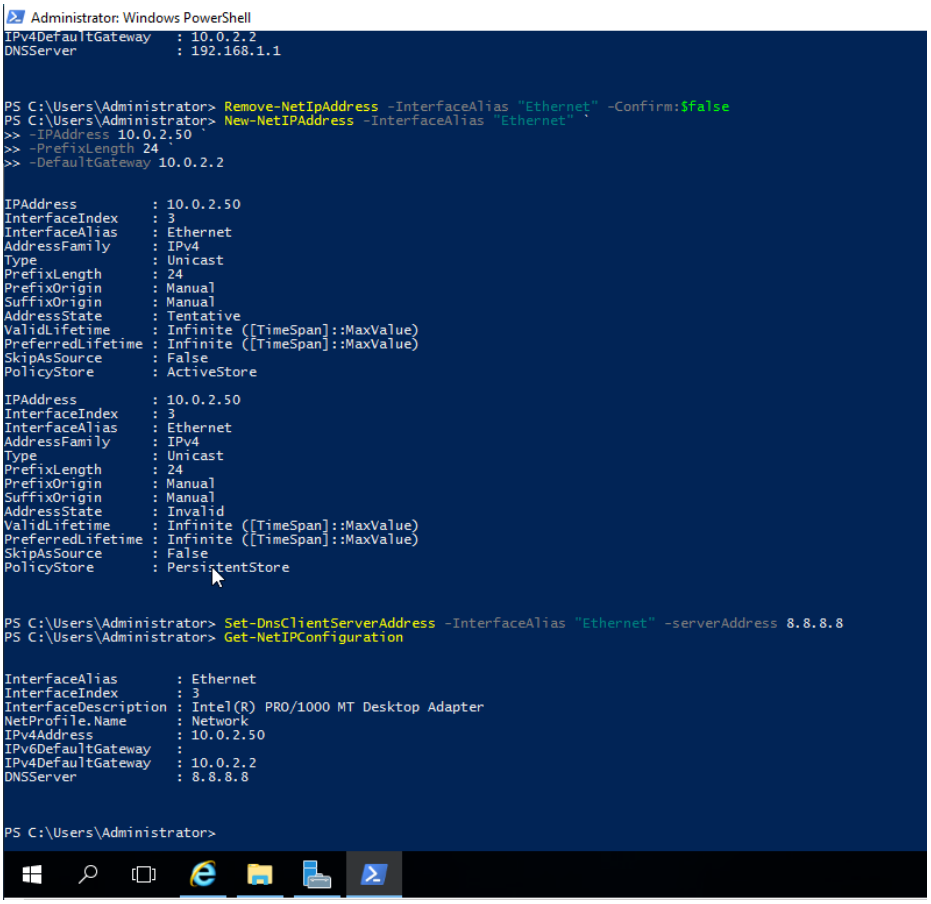


Figure 1. Server Manager during the Add Roles and Features workflow.

Preparation and Initial Conditions:

- The server was already installed and reachable at the local console, so the work began from the Desktop Experience environment with Server Manager available.
- The sequence used in the lab was GUI-based role installation first, followed by PowerShell-based service expansion and network configuration.

TECHNICAL WORK PERFORMED

Server Manager Role and Feature Installation Workflow:

1) Started the Windows Server 2016 Datacenter Desktop Experience VM and logged in as Administrator.
2) Opened Server Manager and selected Manage → Add Roles and Features.
3) Used the default role-based or feature-based installation path on the local server.
4) Advanced through Before You Begin, Installation Type, and Server Selection using the local-target workflow.
5) On the Server Roles page, selected DHCP Server and accepted the DHCP management tools through Add Features.
6) On the Features page, selected Windows Server Backup.
7) A warning appeared indicating that the server did not yet have a static IP address configured for DHCP, but the installation was continued so the role deployment could complete.
8) The results screen confirmed installation succeeded and listed DHCP Server, DHCP Server Tools, and Windows Server Backup as installed.

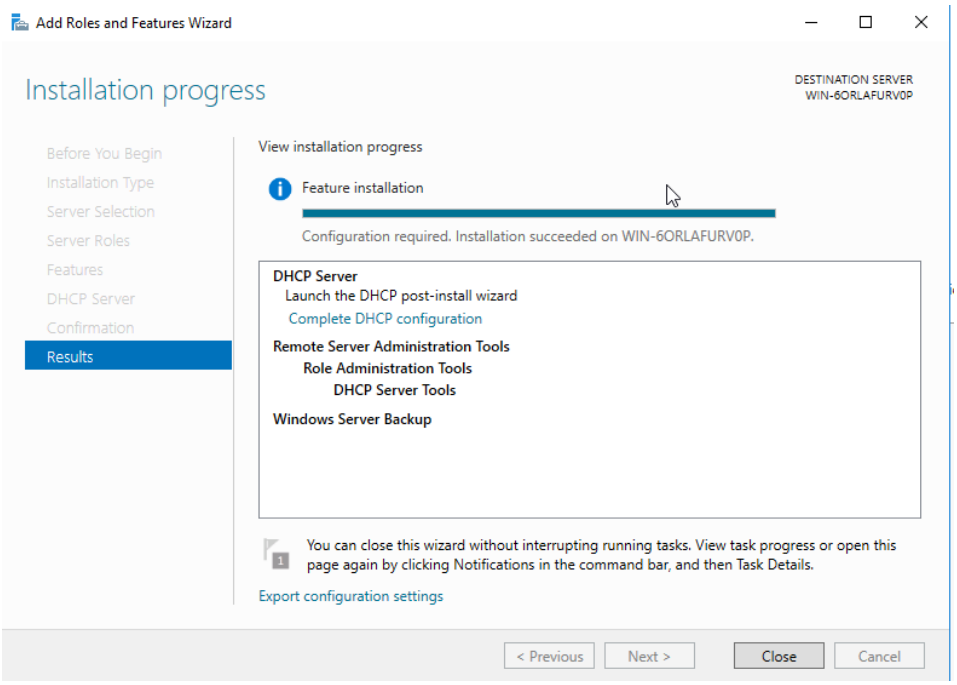


Figure 2. Successful installation of DHCP Server, DHCP Server Tools, and Windows Server Backup.

PowerShell Role Expansion and Local Service Setup:

- After the GUI-based installation completed, Windows PowerShell was used to expand the server role set with additional services.
- The IIS Web Server role was installed with management tools so the machine could host local web content and be checked through a browser test.
- The File Server role was installed to support local sharing and file-service validation.
- A local directory named C:\Shared was created and published as an SMB share so that local file-sharing functionality could be checked from inside the VM.

Local IIS and File-Sharing Checks:

- Internet Explorer was opened inside the server VM and http://localhost was tested.
- The default IIS Windows Server page loaded successfully, confirming that the local web service was active.
- File Explorer was then used to browse the localhost network view and confirm that the Shared folder appeared as a local network share.

KEY COMMANDS, SERVICE CONFIGURATION, AND NETWORK WORK

Key Commands Used During the Lab:

ROLE / FEATURE COMMANDS

```
Get-WindowsFeature
Install-WindowsFeature -Name
DHCP,Windows-Server-Backup
-IncludeManagementTools -Restart
Install-WindowsFeature -Name
Web-Server
-IncludeManagementTools
Install-WindowsFeature -Name
FS-FileServer
```

FILE SHARING COMMANDS

```
New-Item -Path C:\Shared
-ItemType Directory
New-SmbShare -Name Shared -Path
C:\Shared
-FullAccess Everyone
Get-SmbShare
Test-Path \\localhost\Shared
Get-ChildItem \\localhost\Shared
```

NETWORKING COMMANDS

```
Get-NetAdapter
Get-NetIPConfiguration
Remove-NetIPAddress
-InterfaceAlias "Ethernet"
-Confirm:$false
New-NetIPAddress -InterfaceAlias
"Ethernet"
-IPAddress 10.0.2.50
-PrefixLength 24
-DefaultGateway 10.0.2.2
Set-DnsClientServerAddress
-InterfaceAlias "Ethernet"
-ServerAddresses 8.8.8.8
ping 8.8.8.8
ping google.com
```

PowerShell Installation and Share Creation:

- PowerShell was used to install both IIS and File Server directly, which provided a repeatable approach compared to relying only on the GUI wizard.
- The directory C:\Shared was created and then published as the Shared SMB resource.
- The initial share configuration used broad access so that the basic file-service path could be checked quickly before later security tightening.

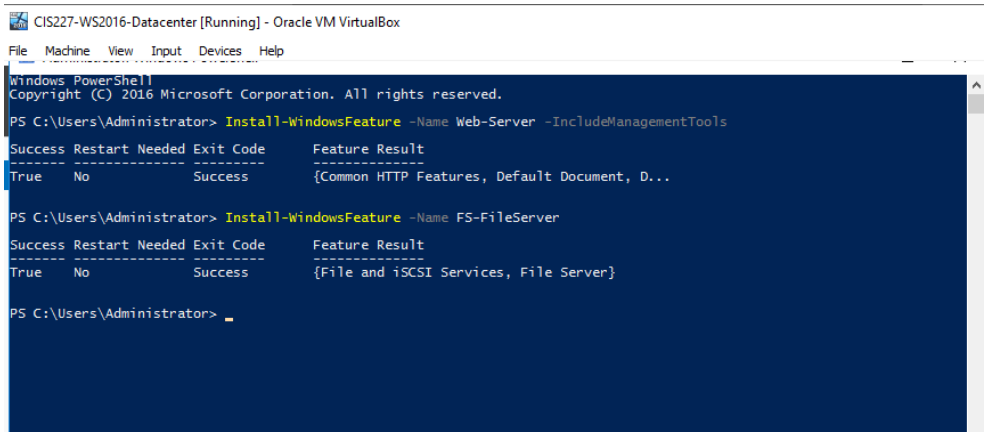


Figure 3. PowerShell installation of IIS Web Server and File Server roles.

Initial Network Discovery and Static IP Reconfiguration:

- PowerShell inspection showed that the active interface was Ethernet and that the server was initially using VirtualBox NAT client addressing.
- The original client configuration used address 10.0.2.15 with default gateway 10.0.2.2.
- The NIC was then reconfigured to use static IPv4 address 10.0.2.50 with default gateway 10.0.2.2 and DNS server 8.8.8.8.

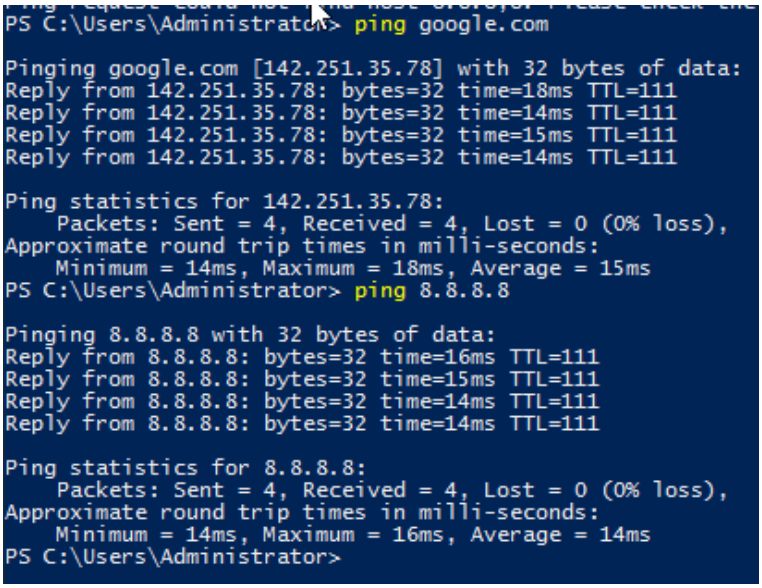
Operational Note: DHCP Role vs DHCP Client Addressing

- Installing the DHCP Server role and using DHCP as a network client are separate conditions on the same machine.
- Changing the NIC to a static address did not remove DHCP Server; it only stopped the server from depending on a changing client lease for its own network identity.

SUMMARY, RESULTS, AND NEXT PRIORITIES

Connectivity and Final Validation:

- After the static IP and DNS changes were completed, two connectivity tests were used to verify that the new configuration was operational.
- Ping 8.8.8.8 succeeded, confirming outbound routing and basic network connectivity.
- Ping google.com also succeeded, confirming that DNS resolution was working correctly after the DNS server change.
- Together with the local IIS and SMB checks, these results confirmed that the system had moved from a basic install state into a functioning Windows Server baseline.



```
PS C:\Users\Administrator> ping google.com

Pinging google.com [142.251.35.78] with 32 bytes of data:
Reply from 142.251.35.78: bytes=32 time=18ms TTL=111
Reply from 142.251.35.78: bytes=32 time=14ms TTL=111
Reply from 142.251.35.78: bytes=32 time=15ms TTL=111
Reply from 142.251.35.78: bytes=32 time=14ms TTL=111

Ping statistics for 142.251.35.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 18ms, Average = 15ms
PS C:\Users\Administrator> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=16ms TTL=111
Reply from 8.8.8.8: bytes=32 time=15ms TTL=111
Reply from 8.8.8.8: bytes=32 time=14ms TTL=111
Reply from 8.8.8.8: bytes=32 time=14ms TTL=111

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 16ms, Average = 14ms
PS C:\Users\Administrator>
```

Figure 4. Successful connectivity and DNS validation using ping 8.8.8.8 and ping google.com.

Completion Summary:

- Installed DHCP Server and Windows Server Backup successfully through Server Manager.
- Installed IIS Web Server and File Server successfully through Windows PowerShell.
- Validated IIS locally through http://localhost and validated local SMB sharing through the Shared folder.
- Reconfigured the NIC from DHCP client addressing to a static IP baseline and verified routing and name resolution after the change.

Key Findings:

1) Role installation alone is not enough; local service validation proves that the work actually succeeded.
2) PowerShell provided a direct way to extend the server build beyond the GUI wizard.
3) Static IP configuration and corrected DNS settings matter immediately once the server begins hosting services such as IIS and file sharing.
4) The first file-share configuration worked functionally but still required tighter permissions for a better security baseline.

Next Priorities:

- Priority 1: Secure the Shared folder by removing broad Everyone access and replacing it with tighter Administrator-only or user-specific permissions.
- Priority 2: Replace the default IIS page with a custom test page and continue web-service validation.
- Priority 3: Configure and test a real DHCP scope in a network layout where the server can lease addresses to separate clients.

Final Notes:

- The completed work covered GUI administration, PowerShell-based service installation, local web and file-service checks, and network correction on the same Windows Server 2016 system.
- The strongest outcome of the lab was not only successful installation, but also successful confirmation that the installed services and updated network settings worked together.